# Differentiation: Optimisation

Questions 1–17

**Question 1.** For each of the following shapes, write an algebraic expression for the area, $A$ and the perimeter, $P$:

a) Square with sides of length $x$

b) Rectangle with sides of length $x$ and $2x$

c) Circle with radius $x$

d) Isosceles triangle with base of $x$ and perpendicular height of $2x$

**Question 2.** For each of the following 3D shapes, write an algebraic expression for the volume, $V$ and the total surface area, $A$:

a) Cube with sides of length $x$

b) Cuboid with sides of length $x, x$ and $3x$

c) Cylinder with radius $x$ and a length $h$

d) Isosceles triangular based prism of length $h$, with the triangle surface having base $x$ and a perpendicular height of $x$

**Question 3.** For each of the following, find:

a) $\frac{dV}{dr}$ for $V = 20r - \frac{1}{4}r^2$

b) $\frac{dP}{dt}$ for $P = \frac{1}{t} + \frac{t}{2}$

c) $\frac{dA}{dr}$ for $A = 2r - \frac{10}{r^2}$

d) $\frac{dC}{dx}$ for $C = \frac{10}{x} + \frac{x^2}{4}$

**Question 4.** Solve each of the following equations, leaving your answer in exact form:

a) $25\pi - 4\pi r^2 = 0$, $r > 0$

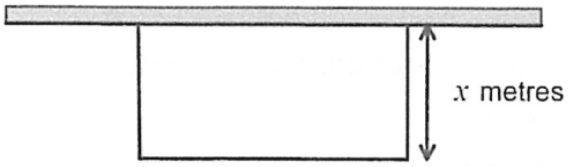
b) $-\frac{144}{x^2} + \frac{1}{4} = 0$, $x > 0$

c) $-\frac{10}{x^2} + \frac{1}{2}x = 0$, $x > 0$

d) $\frac{1}{4}\pi r - \frac{27\pi^2}{r^2} = 0$, $r > 0$

e) $\frac{1}{5}\pi r - \frac{3}{r^2} = 0$, $r > 0$

**Question 5.** A rectangular enclosure is created within a field, with a wall forming one side with a width of $x$ metres. A rope of length 30 metres is used to form the remaining three sides.




a) Show that the area $A$ may be written as $A = 30x - 2x^2$.

b) Hence, find the maximum area of the enclosure.

**Question 6.** The cost of a journey £$C$ when a van when driving at a steady speed in $v$ kilometres per hour may be modelled by

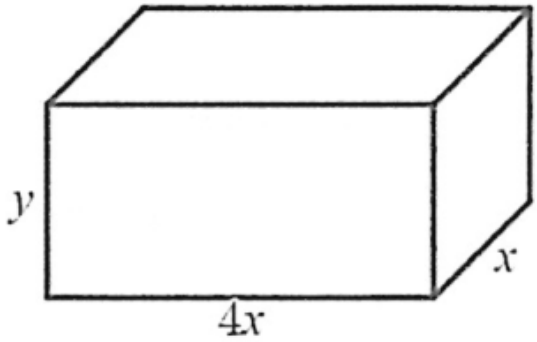
$$C = \frac{1156}{v} + \frac{v}{9} + 50$$

where $v > 0$.

a) Find the speed at which the minimum cost occurs.

b) Find the minimum cost.

**Question 7.** The diagram shows a solid cuboid with lengths $x$ cm, $4x$ cm and $y$ cm.




The total surface area of the cuboid is 100 $\text{cm}^2$.

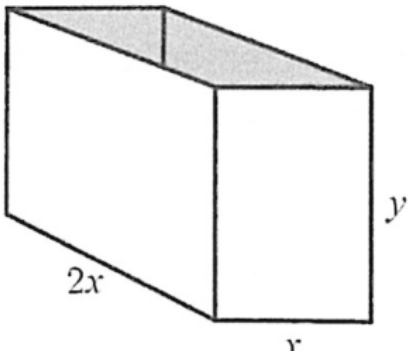
a) Show that the volume may be written as

$$V = 40x - \frac{16}{5}x^3.$$

b) Hence, find the maximum volume of the cuboid giving your answer to 1 decimal place.

c) Find the exact dimensions of the box for this maximum volume to occur.

**Question 8.** The diagram shows an open-topped cuboid tank. Its base has width $x$ metres and length $2x$ metres and has a height of $y$ metres.




It is made from 48 $\text{m}^2$ of sheet metal.

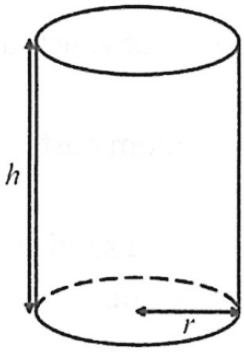
a) Show that the volume may be written as

$$V = 16x - \frac{2}{3}x^3.$$

b) Hence, find the maximum volume giving your answer to 1 decimal place.

c) Find the exact dimensions of the box for this maximum volume to occur.

**Question 9.** The diagram shows a plant pot which is a cylindrical container of radius $r$ cm and height $h$ cm. The plant pot has an open top. The total surface area is $48\pi$ cm$^2$.

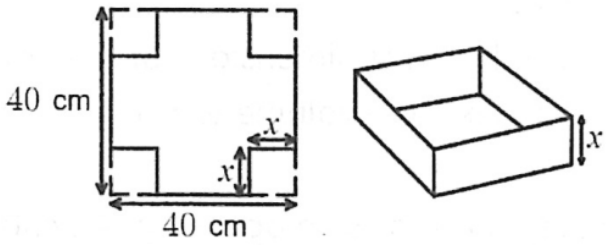



a) Show that the volume may be written as

$$V = 24\pi r - \frac{\pi}{2}r^3.$$

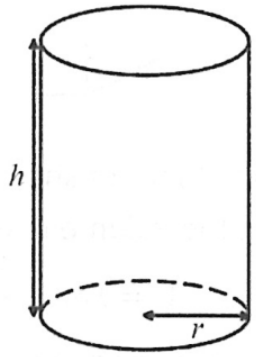
b) Hence find the maximum volume giving your answer in exact form.

**Question 10.** The diagram shows a piece of square cardboard with width 40 cm. Four squares are cut from the corners each of width $x$ cm. The cardboard is then folded into an open tray.




Find the maximum capacity of the tray. Give your answer to 1 decimal place.

**Question 11.** A container is created as a closed cylinder with capacity $432\pi$ cm$^3$.



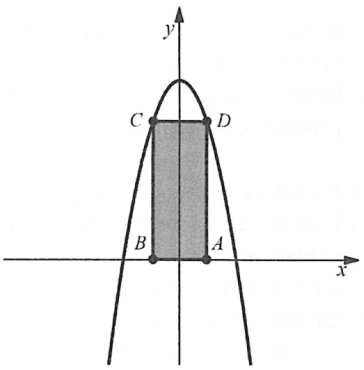

a) Find the dimensions of the container when its surface area is minimised.

b) For sealing the side of the container, a line of fluid is dropped along the length. Find the cost given it costs £0.04 per cm.

**Question 12.** The curve

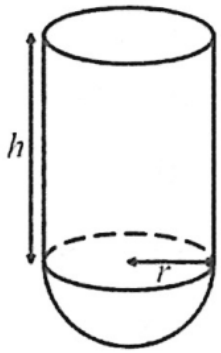
$$f(x) = 10 - x^2$$

and rectangle ABCD is shown in the diagram below.




Given that the rectangle is bounded by the curve and the $x$-axis, find the maximum area that the rectangle can have.

**Question 13.** A container for ice cream desserts is being designed to use as little plastic as possible. It needs to have a volume of $144\pi$ cm$^3$. The vendors of the ice cream would like it to be comprised of an open top semi-sphere with a cylinder with the same radius attached, and that it should have no base or top. The radius of the cylinder is $r$ cm and the length of the cylinder is $h$ cm.




A sphere of radius $r$ has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$.

a) Show that the area of plastic used is given by

$$A = \frac{2}{3}\pi r^2 + \frac{288\pi}{r}.$$

b) Hence find the radius which minimises the area.

c) By calculating the value of $h$ for this to be minimised, state an issue with the vendor's requirement.

**Question 14.** A cylindrical metal container which is closed at either end, has a capacity of 500 cm$^3$. The metal used for the curved side costs 0.03 pence/cm$^2$. The cost for the circular bottom and top costs 0.07 pence/cm$^2$.

a) Show that the cost of the metals for making the container may be given by

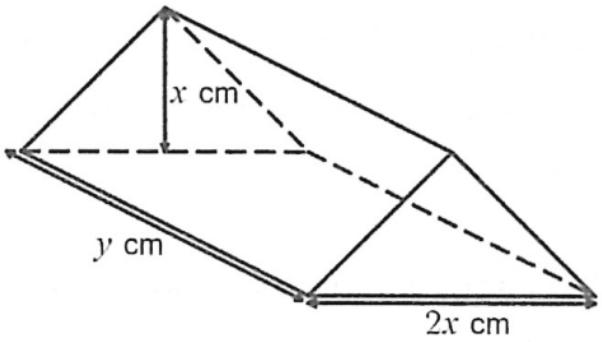
$$C = \frac{7}{50}\pi r^2 + \frac{30}{r}$$

where $r$ is the radius of the container and $C$ is the cost in pence.

b) Find the radius for the minimum cost and height for that cost.

c) Hence calculate the minimum cost of making one of these containers to the nearest pence.

**Question 15.** A solid triangular based prism is made from wood with a volume of $\left(500(\sqrt{2}-1)\right)$ cm$^3$. The two triangular ends are isosceles triangles with base $2x$ cm and perpendicular height $x$ cm. The length of the prism is $y$ cm. The rectangular sides are painted with paint that costs £0.02 per cm$^2$ and the triangular ends are painted with paint that costs £0.04 per cm$^2$.



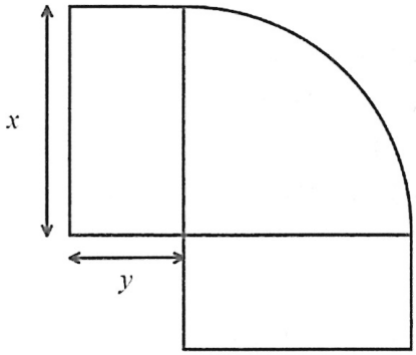

a) Show that the slanted sides (not the base) have a total area of $2\sqrt{2}xy$.

b) Show that the cost of painting the prism is given by

$$C = \frac{20}{x} + \frac{2}{25}x^2.$$

c) Hence calculate the minimum cost of painting one of these prisms.

**Question 16.** A part of a garden is going to be filled with two different types of decorative stones. The design is a quarter circle with radius $x$. Each rectangle has width $y$ metres and length $x$ metres and is laid out as shown in the diagram.



**Question 16 (continued).** For the quarter circle, the stones will be 80% of the cost of the stones for the rectangular areas. The total perimeter is 16 metres.

a) Show that the cost may be given by

$$C = 8px - \left(1 + \frac{\pi}{20}\right) px^2$$

for some cost £$p$ per metre$^2$.

b) Show that the length of $x$ for the maximum cost is occurs when

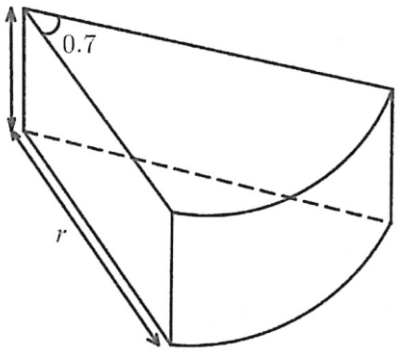
$$x = \frac{80}{20 + \pi}.$$

c) Calculate the area of the quarter circle and the rectangles separately for this value of $x$, giving your answer to 3 significant figures.

**Question 17.**

## Prerequisite knowledge for Q17: Arc Length and Sector Area in Radians

A toy slice of cake is being manufactured and can be modelled by a sector of a circle with radius $r$. The volume is to be 350 cm$^3$. The angle of the sector is 0.7 radians.




a) Show that the surface area may be written in the form

$$A = \frac{2700}{r} + 0.7r^2.$$

b) Hence find the minimum surface area and the values of $r$ and $h$ which obtain this, giving your answer to 3 significant figures.